

1. Let $\mathbf{A}$ and $\mathbf{B}$ both be $n \times n$ and assume that there is an invertible matrix $\mathbf{S}$ such that $\mathbf{SAS}^{-1}$ and $\mathbf{SBS}^{-1}$ are both diagonal (but not necessarily equal). Prove that $\mathbf{AB} = \mathbf{BA}$.

Solution: Let

$$\mathbf{SAS}^{-1} = \mathbf{D}_A \text{ and } \mathbf{SBS}^{-1} = \mathbf{D}_B.$$

Write

$$\mathbf{A} = \mathbf{S}^{-1}\mathbf{D}_A\mathbf{S} \text{ and } \mathbf{B} = \mathbf{S}^{-1}\mathbf{D}_B\mathbf{S}.$$

Note

$$\mathbf{AB} = \mathbf{S}^{-1}\mathbf{D}_A\mathbf{S}\mathbf{S}^{-1}\mathbf{D}_B\mathbf{S} = \mathbf{S}^{-1}\mathbf{D}_A\mathbf{D}_B\mathbf{S} = \mathbf{S}^{-1}\mathbf{D}_B\mathbf{D}_A\mathbf{S} = \mathbf{S}^{-1}\mathbf{D}_B\mathbf{S}\mathbf{S}^{-1}\mathbf{D}_A\mathbf{S} = \mathbf{BA}.$$

2. Suppose that $\mathbf{A}$ is symmetric and positive definite, while $\mathbf{B}$ is symmetric. Prove that there is a real invertible matrix $\mathbf{S}$ such that

$$\mathbf{A} + t\mathbf{B} = \mathbf{S}(\mathbf{I} + t\mathbf{D})\mathbf{S}^T$$

where $\mathbf{D}$ is diagonal, for every t (i.e. none of the matrices in the above expression depends on t).

Solution: Since $\mathbf{A}$ is SPD it has a Cholesky factorization $\mathbf{A} = \mathbf{LL}^T$. Factor out the Cholesky factors from the sum:

$$\mathbf{A} + t\mathbf{B} = \mathbf{L}(\mathbf{I} + t\mathbf{L}^{-1}\mathbf{B}\mathbf{L}^{-T})\mathbf{L}^T.$$

Notice that the matrix $\mathbf{L}^{-1}\mathbf{B}\mathbf{L}^{-T}$ is real and symmetric, so it has a real orthogonal eigenvalue decomposition

$$\mathbf{L}^{-1}\mathbf{B}\mathbf{L}^{-T} = \mathbf{QDQ}^T.$$

Insert this and simplify

$$\mathbf{L}(\mathbf{I} + t\mathbf{QDQ}^T)\mathbf{L}^T = \mathbf{L}(\mathbf{QQ}^T + t\mathbf{QDQ}^T)\mathbf{L}^T = \mathbf{LQ}(\mathbf{I} + t\mathbf{D})(\mathbf{LQ})^T.$$

The matrix $\mathbf{S} = \mathbf{LQ}$ is real and invertible.

3. Suppose that $\mathbf{A}$ and $\mathbf{B}$ are both $n \times n$. Prove that $\mathbf{AB}$ and $\mathbf{BA}$ have the same eigenvalues.

Solution: Suppose that $\mathbf{AB}\vec{v} = \lambda\vec{v}$ with $\vec{v} \neq \vec{0}$. Multiply by $\mathbf{B}$: $\mathbf{BAB}\vec{v} = \lambda\mathbf{B}\vec{v}$. Define $\vec{w} = \mathbf{B}\vec{v}$. Then

$$\mathbf{BA}\vec{w} = \lambda\vec{w}.$$

This equation only implies that λ is an eigenvalue of $\mathbf{BA}$ when $\vec{w} \neq \vec{0}$. So what we've shown so far is

If $\mathbf{AB}$ has eigenvalue λ and eigenvector $\vec{v}$ then $\mathbf{BA}$ also has eigenvalue λ provided that $\mathbf{B}\vec{v} \neq \vec{0}$.

So what happens when $\vec{v} \neq \vec{0}$ but $\mathbf{B}\vec{v} = \vec{0}$? In that case $\lambda = 0$ must be an eigenvalue of $\mathbf{AB}$ because $\mathbf{AB}\vec{v} = \mathbf{A}\vec{0} = \vec{0} = (0)\vec{v}$. So we can conclude at this point that

If $\mathbf{AB}$ has eigenvalue $\lambda \neq 0$ and eigenvector $\vec{v}$ then $\mathbf{BA}$ also has eigenvector $\vec{v}$.

By re-labeling $\mathbf{A}$ and $\mathbf{B}$ we find that

The nonzero eigenvalues of $\mathbf{AB}$ and $\mathbf{BA}$ are the same.

To complete the proof we need to show that if $\lambda = 0$ is an eigenvalue of $\mathbf{AB}$ then it must also be an eigenvalue of $\mathbf{BA}$. Clearly if $\mathbf{A}$ and $\mathbf{B}$ are both invertible then it's not possible for $\lambda = 0$ to be an eigenvalue of $\mathbf{AB}$ or $\mathbf{BA}$. If at least one of the factors is singular then the product is also singular, meaning that it has a zero eigenvalue regardless of the order of the factors. This completes the proof.

4. Suppose that $\mathbf{A}$ and $\mathbf{B}$ are both symmetric and positive definite. Prove that $\mathbf{AB}$ has real, positive eigenvalues. (Hint: Show that $\mathbf{AB}$ is similar to a symmetric and positive definite matrix.)

Solution: Since $\mathbf{A}$ is SPD it has a Cholesky factorization

$$\mathbf{A} = \mathbf{L}\mathbf{L}^T$$

so

$$\mathbf{AB} = \mathbf{L}\mathbf{L}^T\mathbf{B}.$$

The Cholesky factor $\mathbf{L}$ is invertible. Multiply from the left by $\mathbf{L}^{-1}$ and from the right by $\mathbf{L}$ to get

$$\mathbf{L}^{-1}\mathbf{AB}\mathbf{L} = \mathbf{L}^T\mathbf{B}\mathbf{L}.$$

The equation above says that $\mathbf{AB}$ is similar to $\mathbf{L}^T\mathbf{B}\mathbf{L}$, so they have the same eigenvalues. The matrix $\mathbf{L}^T\mathbf{B}\mathbf{L}$ is symmetric, so it has real eigenvalues. Also note that

$$\vec{x}^T\mathbf{L}^T\mathbf{B}\mathbf{L}\vec{x} = \vec{y}^T\mathbf{B}\vec{y} \geq 0 \quad \forall \vec{y} \neq \vec{0}.$$

Since $\vec{y} = \vec{0}$ only when $\vec{x} = \vec{0}$ (because $\mathbf{L}$ is invertible), we have shown that $\mathbf{L}^T\mathbf{B}\mathbf{L}$ is positive definite, and therefore has real eigenvalues.

5. Suppose that $\mathbf{L} \in \mathbb{R}^{n \times n}$ is a symmetric matrix with eigenvalues $\lambda_1, \dots, \lambda_m$. (Note that m can be $\leq n$.) Prove that

$$\mathbf{L} = \lambda_1\mathbf{P}_1 + \dots + \lambda_m\mathbf{P}_m$$

where $\mathbf{P}_i$ is an orthogonal projection matrix that projects orthogonally onto the nullspace of $\mathbf{L} - \lambda_i\mathbf{I}$.

Solution: Since $\mathbf{L}$ is symmetric it has a real orthogonal eigenvalue decomposition

$$\mathbf{L} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T.$$

Write

$$\mathbf{Q} = [\vec{q}_1, \dots, \vec{q}_n]. \quad \mathbf{\Lambda} = \text{diag}(\mu_1, \dots, \mu_n).$$

The eigenvalues of $\mathbf{L}$ are $\mu_1, \dots, \mu_n$. But we already said that the eigenvalues of $\mathbf{L}$ are $\lambda_1, \dots, \lambda_m$! What gives? The list of μ_i has all the eigenvalues including potential repeats, while the list $\lambda_1, \dots, \lambda_m$ has all the eigenvalues *not* counting repeats.

The eigenvalue decomposition can be equivalently written

$$\mathbf{L} = \mu_1 \vec{q}_1 \vec{q}_1^T + \dots + \mu_n \vec{q}_n \vec{q}_n^T.$$

If we group the eigenvalues we get

$$\mathbf{L} = \lambda_1 [\vec{q}_1, \dots] [\vec{q}_1, \dots]^T + \dots + \lambda_m [\vec{q}_r, \dots] [\vec{q}_r, \dots]^T.$$

Each of the matrices $[\vec{q}_r, \dots] [\vec{q}_r, \dots]^T$ is an orthogonal projection matrix that projects orthogonally onto the eigenspace associated with λ_r . (The subscript r is not a typo – it reflects the fact that $\vec{q}_r$ is an eigenvector for μ_r , and μ_r equals λ_m . The use of r is in no way related to rank.) The notation in this solution is not great, but making it precisely correct requires an overload of notation.

6. Prove that if $\mathbf{A}$ is invertible (and, for simplicity, real), then there is an orthogonal matrix $\mathbf{U}$ and a symmetric and positive definite matrix $\mathbf{E}$ such that $\mathbf{A} = \mathbf{U}\mathbf{E}$. This is analogous to writing complex numbers in polar form $x + iy = \rho e^{i\theta}$. Show that if $\mathbf{A}$ is square but not invertible then there is a similar factorization but with $\mathbf{E}$ being symmetric and non-negative definite.

Solution: Consider the SVD of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T = \mathbf{P}\mathbf{Q}^T\mathbf{Q}\mathbf{\Sigma}\mathbf{Q}^T = \mathbf{U}\mathbf{E}$$

where

$$\mathbf{U} = \mathbf{P}\mathbf{Q}^T \text{ and } \mathbf{E} = \mathbf{Q}\mathbf{\Sigma}\mathbf{Q}^T.$$

The diagonal entries of $\mathbf{\Sigma}$ are the eigenvalues of $\mathbf{E}$. When $\mathbf{A}$ is invertible $\mathbf{\Sigma}$ has positive diagonal entries, and when $\mathbf{A}$ is not invertible $\mathbf{\Sigma}$ has some positive and some zero diagonal entries. The matrix $\mathbf{U}$ is a product of orthogonal matrices and is therefore orthogonal.

7. Let $\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}$ be diagonalizable. Find a symmetric positive definite matrix $\mathbf{K}$ that defines an inner product $\langle \vec{x}, \vec{y} \rangle = \vec{x}^T \mathbf{K} \vec{y}$ such that $\mathbf{A}$ commutes with its $\mathbf{K}$ -adjoint.

Solution: First recall that the $\mathbf{K}$ -adjoint of $\mathbf{A}$ is

$$\mathbf{A}^\dagger = \mathbf{K}^{-1} \mathbf{A}^T \mathbf{K}.$$

We will draw inspiration from the idea that ‘commutes with adjoint’ is a generalization of the definition of a normal matrix, and that the eigenvectors of normal matrices are orthogonal. So we want to find an SPD matrix $\mathbf{K}$ such that the Gram matrix formed using the eigenvectors of $\mathbf{A}$ and the inner product defined by $\mathbf{K}$ is the identity:

$$\mathbf{S}^T \mathbf{K} \mathbf{S} = \mathbf{I}.$$

Consider the SVD of $\mathbf{S}$:

$$\mathbf{S} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T$$

where all factors are square and invertible. If we set

$$\mathbf{K} = \mathbf{P}\mathbf{\Sigma}^{-2}\mathbf{P}^T$$

then we find

$$\mathbf{S}^T \mathbf{K} \mathbf{S} = \mathbf{Q} \mathbf{\Sigma} \mathbf{P}^T \mathbf{P} \mathbf{\Sigma}^{-2} \mathbf{P}^T \mathbf{P} \mathbf{\Sigma} \mathbf{Q}^T = \mathbf{I},$$

as desired.

We should check that our intuition has guided us correctly though, i.e. does $\mathbf{A}$ commute with its $\mathbf{K}$ -adjoint? First notice that the above equation implies

$$\mathbf{S}^{-1} \mathbf{K}^{-1} \mathbf{S}^{-T} = \mathbf{I}$$

which implies

$$\mathbf{K}^{-1} \mathbf{S}^{-T} = \mathbf{S}.$$

We also have

$$\mathbf{S}^T \mathbf{K} = \mathbf{S}^{-1}.$$

Now simplify

$$\mathbf{A}^\dagger \mathbf{A} = \mathbf{K}^{-1} \mathbf{A}^T \mathbf{K} \mathbf{A} = \mathbf{K}^{-1} \mathbf{S}^{-T} \mathbf{\Lambda} \mathbf{S}^T \mathbf{K} \mathbf{S} \mathbf{\Lambda} \mathbf{S}^{-1} = \mathbf{K}^{-1} \mathbf{S}^{-T} \mathbf{\Lambda}^2 \mathbf{S}^{-1} = \mathbf{S} \mathbf{\Lambda}^2 \mathbf{S}^{-1} = \mathbf{A}^2$$

and

$$\mathbf{A} \mathbf{A}^\dagger = \mathbf{A} \mathbf{K}^{-1} \mathbf{A}^T \mathbf{K} = \mathbf{S} \mathbf{\Lambda} \mathbf{S}^{-1} \mathbf{K}^{-1} \mathbf{S}^{-T} \mathbf{\Lambda} \mathbf{S}^T \mathbf{K} = \mathbf{S} \mathbf{\Lambda}^2 \mathbf{S}^T \mathbf{K} = \mathbf{S} \mathbf{\Lambda}^2 \mathbf{S}^{-1} = \mathbf{A}^2.$$

We have therefore found that every diagonalizable matrix is normal with respect to some inner product.

8. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have full column rank. Prove that $\|\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}\|_2 = 1/\sigma_n$ where σ_n is the n^{th} singular value of $\mathbf{A}$. (Note that $\mathbf{A}^{-1}$ is not defined, so your proof cannot use $\mathbf{A}^{-1}$.)

Solution: Consider the SVD of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T$$

where $\mathbf{Q}$ is orthogonal, $\mathbf{\Sigma}$ is diagonal and invertible, and $\mathbf{P}$ is non-square. Plug in and simplify:

$$\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T(\mathbf{Q}\mathbf{\Sigma}^T\mathbf{P}^T\mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T)^{-1} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T(\mathbf{Q}\mathbf{\Sigma}^2\mathbf{Q}^T)^{-1} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^T\mathbf{Q}\mathbf{\Sigma}^{-2}\mathbf{Q}^T = \mathbf{P}\mathbf{\Sigma}^{-1}\mathbf{Q}^T.$$

This is an SVD¹ of $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$. The singular values of $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ are one over the singular values of $\mathbf{A}$. The 2-norm of $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ is the largest singular value of $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$, which is one over the smallest singular value of $\mathbf{A}$, which is $1/\sigma_n$.

¹Technically we need to insert permutation matrices to re-order the singular values so that they decrease along the diagonal.